

Grieco, McDevitt (2016) Summary

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Broad Question

Broadly, Grieco and McDevitt seek to quantify the tradeoff providers face between increasing their capacity for patients and improving quality of care. The clear trade off is that without increasing both simultaneously, you are either losing relative quality via bringing in too many patients or “wasting” your treatment’s quality by not having enough patients.

Specific Question

Using dialysis treatment centers as a vehicle, the authors analyze the quality-quantity tradeoff. These centers possess many valuable qualities (easily estimable production function, observe input-output, observable differences in outcomes, uniform payments for treatment) which make them desirable for analysis. They will use estimation of center production functions to ascertain optimal level of quality at the firm level.

Policy Implication

Dialysis is a huge chunk of MediCare spending (6%) and thus, this is an inherently relevant paper. The authors even point this out in their introduction. Thus, the findings of the authors could contribute meaningfully to policy for funding related to productivity in healthcare. For instance, they find a 1.6% increase in patient quantity is achieved by a decrease in quantity (measured by increase in septic infection rate of 1%). Thus, using results along these lines can help to incentivize quality spending along with quantity.

Economic Implication

Economically, this is interesting mostly for the setting. The dialysis treatment industry has many qualities which allow for easy estimation of firm-level effects. Further, there is a novelty to their application of production functions to healthcare which is useful for further iteration.

What We Learn

Firstly, they interestingly find decreasing returns to scale, suggesting there is likely OVB. They find that, in the main tradeoff, a 1% increase in quality $\rightarrow$ 1.55% reduction in patient load or an increase in workforce by 6.4%. To put things monetarily, a one infection decrease per year $\approx$ \$75,000 in opportunity cost. Using estimates in the literature, they find that social welfare is increased in non-hospitalization cost of infection exceeds \$50,000, but reduces social welfare if the cost is less than that amount, aiding policy makers.

Empirics

Model of production. Honestly, seems like they did not specify it as well as they could have, something they discuss at multiple points. There are many estimates recovered which are weird at best or violate the model at worst. I think the idea is great, but maybe another swing at execution would be fruitful.